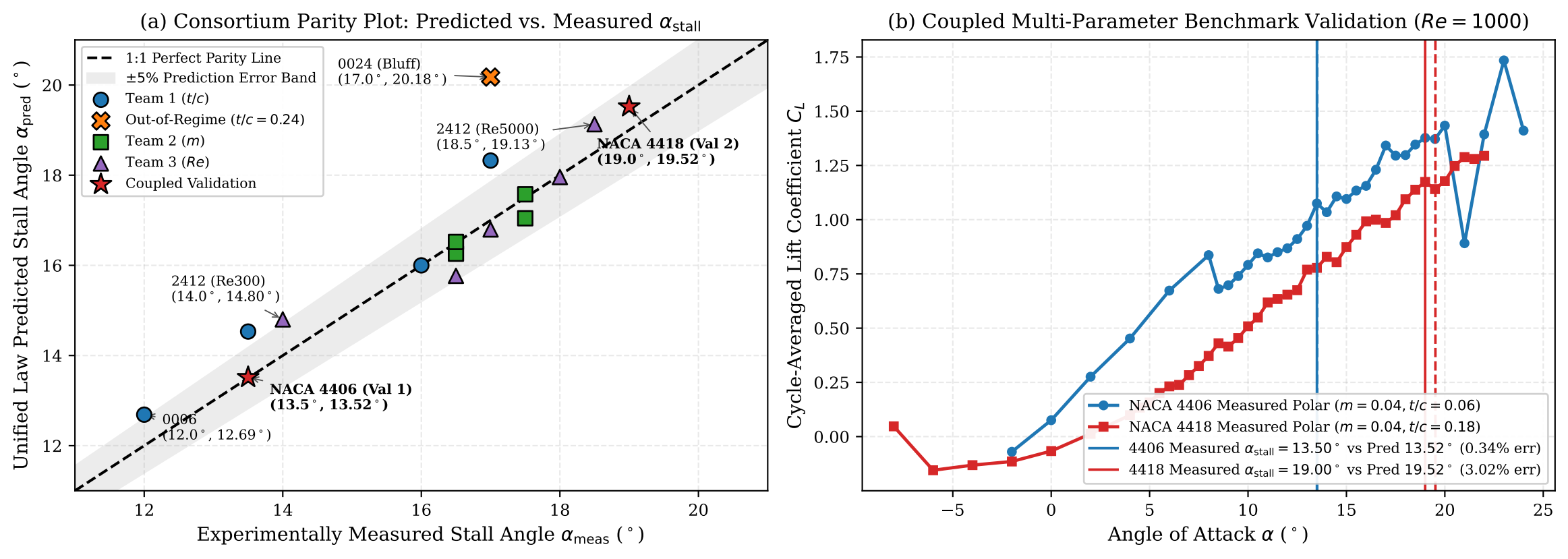
(a) Consortium Parity Plot: Predicted vs. Measured α_{stall}
1:1 Perfect Parity Line
±5% Prediction Error Band
Team 1 (t/c)
Out-of-Regime ($t/c = 0.24$)
Team 2 (m)
Team 3 (Re)
Coupled Validation
0024 (Bluff)
(17.0°, 20.18°)
2412 (Re5000)
(18.5°, 19.13°)
NACA 4418 (Val 2)
(19.0°, 19.52°)
2412 (Re300)
(14.0°, 14.80°)
NACA 4406 (Val 1)
(13.5°, 13.52°)
0006
(12.0°, 12.69°)
Unified Law Predicted Stall Angle α_{pred} (°)
Experimentally Measured Stall Angle α_{meas} (°)
(b) Coupled Multi-Parameter Benchmark Validation ($Re = 1000$)
Cycle-Averaged Lift Coefficient C_L
Angle of Attack α (°)
NACA 4406 Measured Polar ($m = 0.04, t/c = 0.06$)
NACA 4418 Measured Polar ($m = 0.04, t/c = 0.18$)
4406 Measured $\alpha_{stall} = 13.50°$ vs Pred 13.52° (0.34% err)
4418 Measured $\alpha_{stall} = 19.00°$ vs Pred 19.52° (3.02% err)